

Functional dependency

Dr. Gunjan Chugh
Assistant Professor
DTU

Functional dependency

- A functional dependency (FD) in a Database Management System (DBMS) is a **constraint between two sets of attributes in a table, where one set of attributes uniquely determines the value of another set of attributes.**
- This relationship is denoted as

$$X \rightarrow Y$$

which means "X determines Y" or "Y is functionally dependent on X".

Core Concepts

For any Functional Dependency: $X \rightarrow Y$

- **Determinant (LHS):** The attribute or set of attributes on the left-hand side **(X)** of the arrow. Its value determines the value of the other attribute(s).
- **Dependent (RHS):** The attribute or set of attributes on the right-hand side **(Y)** of the arrow, whose value is determined by the determinant.
- **Formal Definition:** For any two rows (tuples) in a table, if they have the same value for attribute set X, they must also have the same value for attribute set Y.

Example

RollNo	Name	Dept
101	Amit	CSE
102	Neha	IT
103	Rahul	CSE

Functional Dependencies:

RollNo $\rightarrow$ *Name*

RollNo $\rightarrow$ *Dept*

Because RollNo uniquely identifies each student.

Types of Functional dependencies

1. Trivial Functional Dependency

- In Trivial Functional Dependency, a dependent is always a subset of the determinant.
- For example:
 - If $X \rightarrow Y$ and Y is the subset of X , then it is called trivial functional dependency.

Symbolically: $A \rightarrow B$ is trivial functional dependency if B is a subset of A .

- The following dependencies are also trivial: $A \rightarrow A$ & $B \rightarrow B$

2. Non-trivial Functional Dependency

- In **Non-trivial functional dependency**, the dependent is strictly not a subset of the determinant.
- For example, **If $X \rightarrow Y$ and Y is not a subset of X** , then it is called Non-trivial functional dependency.

Example 1 :

- Id $\rightarrow$ Name
- Name $\rightarrow$ DOB

3. Semi Non-Trivial Functional Dependency

A functional dependency:

$$X \rightarrow Y$$

is called **semi non-trivial** if:

$$X \cap Y \neq \emptyset$$

i.e., **some attributes of Y are also present in X, but Y is not completely contained in X.**

Example:

$$A, B \rightarrow B, C$$

- Here, attribute **B** is common in both sides, so it is semi non-trivial.

4. Transitive Functional Dependency

- In transitive functional dependency, dependent is indirectly dependent on determinant.
- i.e.

If $a \rightarrow b$ & $b \rightarrow c$

then

$a \rightarrow c$

This is a transitive functional dependency.

Example: Transitive FD

RollNo	DeptID	DeptName
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FDs:

$$\begin{aligned} RollNo &\rightarrow DeptID \\ DeptID &\rightarrow DeptName \end{aligned}$$

So:

$$RollNo \rightarrow DeptName$$

This is **Transitive Dependency**.

5. Full Functional Dependency (Full FD)

- A functional dependency:

$$X \rightarrow Y$$

- is called a **Full Functional Dependency** if:

Y depends on the **entire** determinant X

- Y does **not** depend on any proper subset of X

In simple words:

You cannot remove any attribute from X and still determine Y.

Example of Full FD

- Table: StudentCourse(StudentID,CourseID,Grade)

Candidate Key = (StudentID, CourseID)

- FD:

StudentID, CourseID → Grade

Why above is Full FD?

- Grade depends on both StudentID and CourseID
- StudentID alone cannot determine Grade
- CourseID alone cannot determine Grade
- So this is a **Full Functional Dependency**.

7. Partial Functional Dependency (Partial FD)

A functional dependency:

$$X \rightarrow Y$$

is called a **Partial Dependency** if:

- X is a **composite key**
- Y depends only on a **part** of X, not the whole key
- In simple words:
- **Y can be determined by removing some attribute from X.**

Example of Partial FD

- Table: StudentCourse(StudentID, CourseID, StudentName)

Candidate Key = (StudentID, CourseID)

- FD:

StudentID, CourseID → StudentName

- But actually:

StudentID → StudentName

- So StudentName depends only on StudentID (part of key).
- Hence, it is a **Partial Dependency**

Prime & Non-Prime Attribute

Prime Attribute

- An attribute is called a **Prime Attribute** if it is part of **any candidate key**.

Non-Prime Attribute

- An attribute is called a **Non-Prime Attribute** if it is **not part of any candidate key**.

Example 1

- Relation: $R(A, B, C, D)$
- Functional Dependencies:

$$A \rightarrow C$$

- Candidate Key: A, B
- So **Prime Attributes are: A,B**
- Because they are part of candidate key.

Example 2 (Multiple Candidate Keys)

- Relation: $R(StudentId, Email, Phone, Name)$
- Candidate Keys:
 - StudentID
 - Email
- Prime Attributes:
 - StudentID
 - Email
- Because both are part of some candidate key.

Example 3

- Relation: $R(A, B, C, D)$
- Candidate Key: A, B
- So Non-Prime Attributes: C, D
- Because they are not part of any candidate key

Example 4

- Relation: $R(A, B, C, D, E)$
- FDs:

$$\begin{array}{l} A \rightarrow C \\ B \rightarrow D \end{array}$$

- Candidate Key:

$$A, B$$

- Prime Attributes:

$$A, B$$

- Non-Prime Attributes:

$$C, D, E$$

Attribute Closure

- Functional dependency and attribute closure are essential for maintaining data integrity and building effective, organized and normalized databases.
- **Attribute closure of an attribute set can be defined as set of attributes which can be functionally determined from it.**
- **The closure of a set of attributes X , written as:**
 X^+

is the set of all attributes that can be functionally determined from X using the given functional dependencies.

- **X^+ contains X itself and all attributes that X can derive using FDs.**

How to find attribute closure of an attribute set?

To find attribute closure of an attribute set:

- Add elements of attribute set to the result set.
- Recursively add elements to the result set which can be functionally determined from the elements of the result set.

Candidate Key

- A candidate key is the **minimal attribute set** whose closure gives all attributes of the relation.
- $X^+ = R$
- and X is minimal $\rightarrow X$ is a candidate key.

QUESTION

- Consider a Relation $R(A,B,C,D,E)$
- FD's

$$\begin{aligned}A &\rightarrow C \\ B &\rightarrow D \\ CD &\rightarrow E\end{aligned}$$

- Find
 - (1.) $(AB)^+$
 - (2.) Super Key
 - (3.) Candidate Key

Steps to Find Candidate Key Using Attribute Closure Method

1. Write the relation with all its attributes.
2. List all given functional dependencies (FDs).
3. Identify attributes that never appear on the RHS of any FD; these attributes must be part of every candidate key.
4. Select a set of attributes and compute its attribute closure (X^+).
5. Apply functional dependencies repeatedly until no new attributes can be added to the closure.
6. If the closure (X^+) contains all attributes of the relation, then X is a super key.
7. Check minimality by removing attributes one by one from X.
If closure is no longer complete after removal, then the key is minimal.
8. The minimal super key obtained is the candidate key.

How to Find Candidate Key Using Closure Method

Step 1: Write the Relation

$R(A,B,C,D,E)$

Step 2: Write the Functional Dependencies

$A \rightarrow C$

$B \rightarrow D$

$CD \rightarrow E$

Step 3: Find Attributes That Must Be in Key

- Any attribute that never appears on the RHS must be part of every candidate key.
- RHS attributes: C,D,E
- So attributes not on RHS: A, B
- Thus key must include: A, B

- **Step 4: Compute Closure of AB**

Start with:

$$(AB)^+ = \{A, B\}$$

Apply FD 1:

$$A \rightarrow C$$

Add C:

$$(AB)^+ = \{A, B, C\}$$

Apply FD 2:

$$B \rightarrow D$$

- Add D:

$$(AB)^+ = \{A, B, C, D\}$$

Apply FD 3:

$$CD \rightarrow E$$

- Since C and D are present, add E:

$$(AB)^+ = \{A, B, C, D, E\}$$

Step 5: Check Closure

$$(AB)^+ = \text{All attributes of } R$$

- **So AB is a Super Key.**

Step 6: Check Minimality

- Remove A:

$$B^+ = \{B, D\}$$

- Not all attributes
- Remove B:

$$A^+ = \{A, C\}$$

- Not all attributes

So AB is minimal.

- Therefore, **Candidate Key=AB**

Practice Question 1

- Relation: $R(A, B, C, D)$
- FDs:

$$A \rightarrow B$$
$$B \rightarrow C$$

Tasks:

1. Find A^+
2. Find Candidate Key(s)

Answer

- $A^+ = \{A, B, C\}$
- Since D is missing, A alone is not key.
- Candidate Key: AD

Practice Question 2

- Relation: $R(A, B, C, D, E)$
- FDs:

$$\begin{aligned}A &\rightarrow B \\ B &\rightarrow C \\ AC &\rightarrow D \\ D &\rightarrow E\end{aligned}$$

- **Tasks:** Find Candidate Key(s)

Practice Question 3

- Relation: $R(A, B, C, D, E)$
- FDs:

$$AB \rightarrow C$$

$$C \rightarrow D$$

$$D \rightarrow E$$

Tasks: Find Candidate Key(s)

Practice Question 4

- Relation: $R(A,B,C,D,E,F)$
- FDs:

$$A \rightarrow BC$$

$$CD \rightarrow E$$

$$B \rightarrow D$$

$$E \rightarrow F$$

- **Tasks:** Find Candidate Key(s)